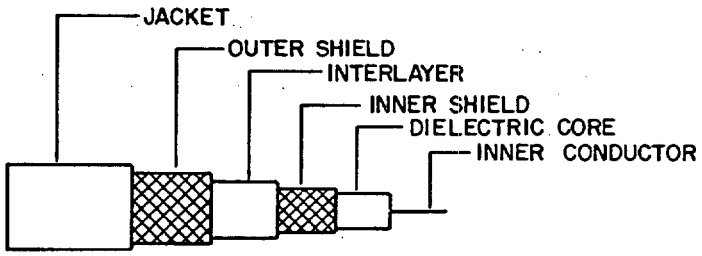
MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, TRIAXIAL, DIAMETER 0.500 INCH, 50 OHM, WATERBLOCKED AND NON-WATER BLOCKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-17.



CABLE SHOULD BE CONSTRUCTED OF NON-TOXIC MATERIAL

FIGURE 1. Triaxial cable, radio frequency M17/135-00001 (water blocked).

TABLE I. Description.

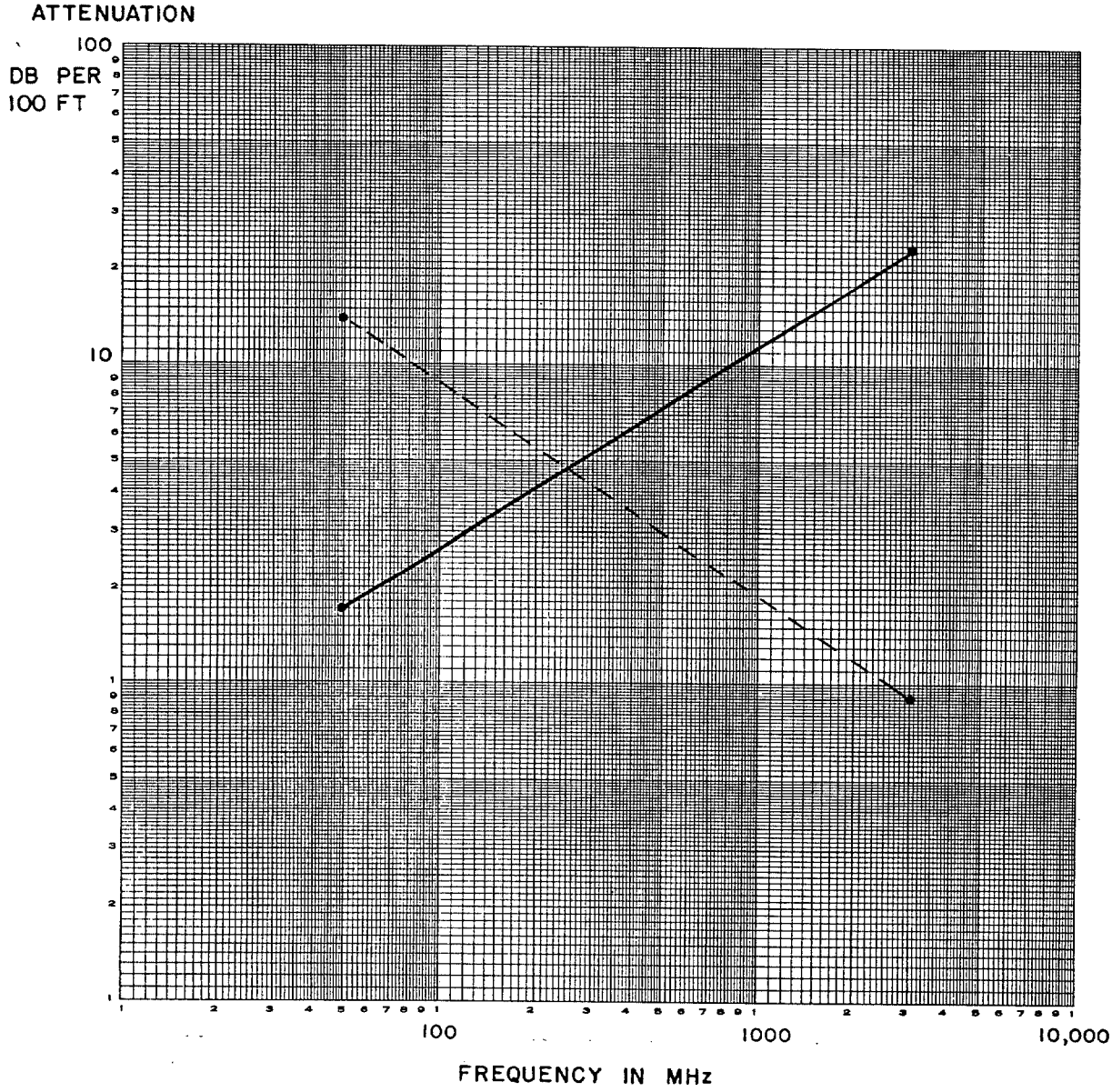
Components	Construction
Inner conductor	7 strands of silver-coated copper wire each strand 0.0296 inch diameter Diameter: 0.089 inch ±0.001
Dielectric core	Type A-1: Solid polyethylene Diameter: 0.285 inch ±0.010
Inner shield	Single braid of AWG #33 silver-coated copper wire Diameter: 0.340 inch maximum Coverage: 90% minimum Carriers: 24 Ends: 7 Picks/inch: 8 Separator tape optional
Interlayer	Type A-1: Solid polyethylene Diameter: 0.365 inch ±0.010
Outer shield	Single braid AWG #33 silver-coated copper wire Diameter: 0.414 inch maximum Coverage: 85% minimum Carriers: 24 Ends: 8 Picks/inch: 5.11 Separator tape optional
Jacket	Type XII: Polyurethane Diameter: 0.500 inch ±0.010

NOTE: (Free strippable blocking compound needed where necessary to meet hydrostatic test. The inner shield may be partially imbedded in the dielectric. Applicable to M17/135-00001 only).

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Maximum attenuation ————————

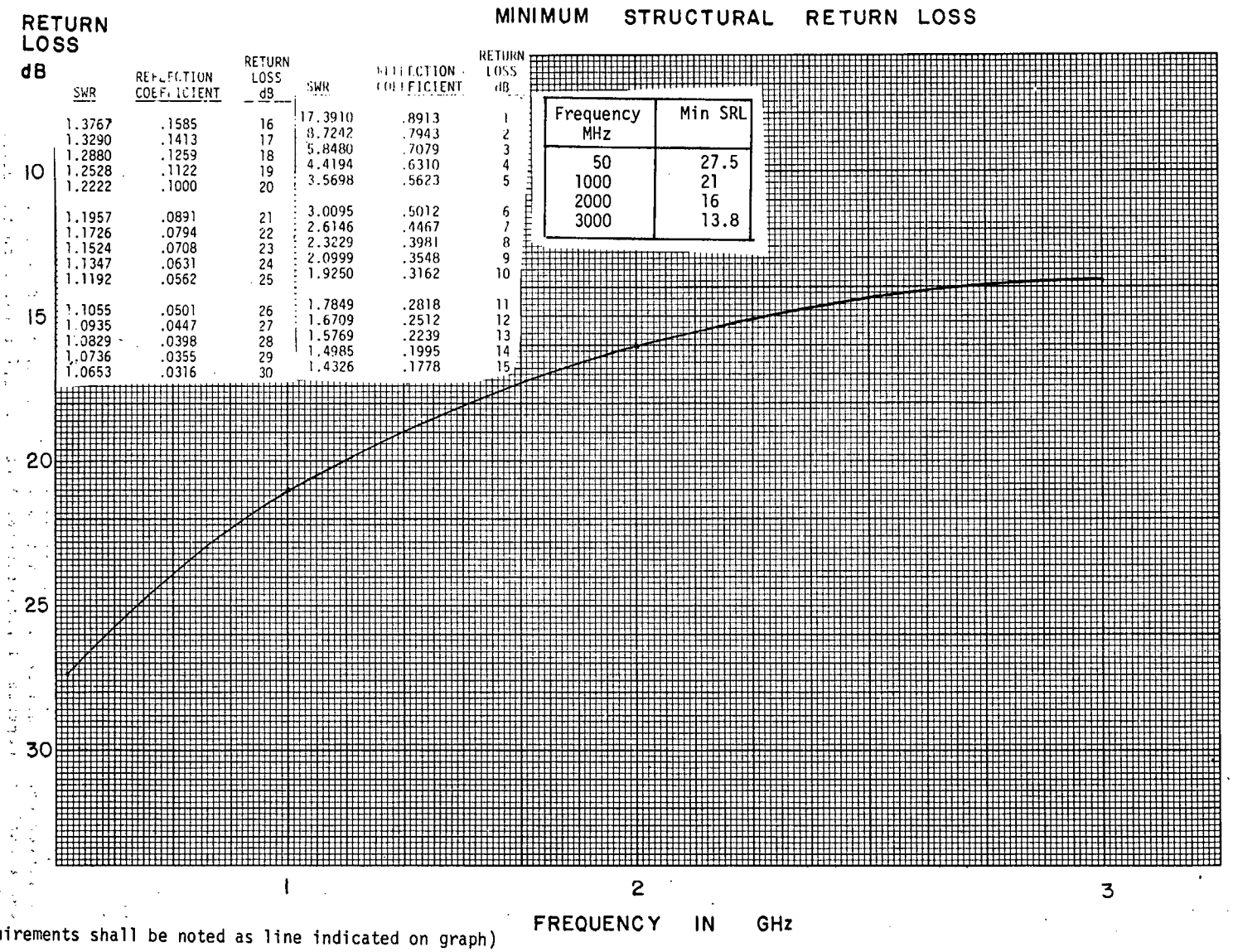
Maximum power — — — — at 25°C sea level

NOTE: (Test requirements shall be noted as line indicated on graph).

Frequency MHz	Attenuation dB	Watts
50	1.7	1400
100	2.5	850
400	6	350
1000	11	190
3000	22	90

FIGURE 2. Power rating and attenuation.

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SWR	REFLECTION COEFFICIENT	RETURN LOSS dB	SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
1.3767	.1585	16	17.3910	.8913	1
1.3290	.1413	17	8.7242	.7943	2
1.2880	.1259	18	5.8480	.7079	3
1.2528	.1122	19	4.4194	.6310	4
1.2222	.1000	20	3.5698	.5623	5
1.1957	.0891	21	3.0095	.5012	6
1.1726	.0794	22	2.6146	.4467	7
1.1524	.0708	23	2.3229	.3981	8
1.1347	.0631	24	2.0999	.3548	9
1.1192	.0562	25	1.9250	.3162	10
1.1055	.0501	26	1.7849	.2818	11
1.0935	.0447	27	1.6709	.2512	12
1.0829	.0398	28	1.5769	.2239	13
1.0736	.0355	29	1.4985	.1995	14
1.0653	.0316	30	1.4326	.1778	15

(Test requirements shall be noted as line indicated on graph)

FIGURE 3. Structural return loss.

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ENGINEERING INFORMATION:

Continuous working voltage: 3,700 Vrms, maximum.

Operating frequency: 3 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40° to 70°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.173 ohm per 100 feet.

Elongation: 30 percent, minimum.

Engineering notes: This cable useful for nonhosing low temperature applications.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Eccentricity: 10 percent, maximum.

Adhesion of conductors: M17/135-0001 7 pounds minimum, 65 pounds maximum.
M17/135-0002 7 pounds minimum, 50 pounds maximum.

Aging stability: +98° ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Dimensional stability:

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 16 lbs per 100 feet, maximum.

Watertightness (Not applicable to M17/135-0002): One end of a five foot length of completed cable shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing gland in the terminal fitting shall be a metal alloy having a maximum melting point of 88°C. The specimen shall be subjected to a water pressure of 25 lb_f/in^2 for a period of 6 hours. Any evidence of water leakage shall be a cause for rejection.

Hydrostatic (open end) (Not applicable to M17/135-0002): A 5-foot specimen of completed cable shall be fitted into a gland similar to USL 44500B-45. The cable shall be capable of withstanding 1,000 lb_f/in^2 applied to the gland end of the specimen for a period of 2 hours. Any evidence of water leakage shall be cause for rejection.

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Electrical:

Test frequency: 50 MHz to 3 GHz.

Spark test: 8,000 Vrms, mimimum.

Voltage withstanding: 10,000 Vrms, minimum.

Insulation resistance: Not applicable.

Corona extinction voltage: 5,000 Vrms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

Contamination: Not applicable.

NOTES: The impedance of the inner conductor to the outer shield shall not vary more than 10 percent from 1 MHz to 200 MHz at operating temperature. The outer shield shall have an attenuation of 70 dB from 1 MHz to 500 MHz per MIL-STD-461, method RS02 and RS03.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation	Type
M17/135-00001	TRF-8	Water blocked
M17/135-00002	YR19211	Non-water blocked

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 17, 85, 99
DLA - IS

User activities:
Army - AV
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-C-17/135A

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

HQ, USA CORADCOM
ATTN: DRDCO-CM-DM
Fort Monmouth, NJ 07703

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C. REASON FOR RECOMMENDED CHANGE(S)

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